

## Solute, Solvent and Solution

A solution is a uniform mixture — the solute dissolves into the solvent and spreads evenly throughout.

### The three words to know

A **solution** is a **uniform mixture** (like salt or sugar in water — every sip tastes the same).

| Term            | Meaning                                                 |
|-----------------|---------------------------------------------------------|
| <b>Solute</b>   | The substance that dissolves (e.g. salt, sugar)         |
| <b>Solvent</b>  | The substance it dissolves in (e.g. water)              |
| <b>Solution</b> | The uniform mixture formed: Solute + Solvent → Solution |

- When a **solid dissolves in a liquid**, the solid is the solute and the liquid is the solvent.
- When **two liquids** mix, the one present in the **smaller amount** is the solute, the one in the **larger amount** is the solvent.

**Gulab jamun syrup:** even though there's a lot of sugar in a little water, **water is still the solvent** and sugar the solute — because sugar is the one that dissolves.

**Water** dissolves so many substances that it is called a good (almost "universal") solvent. Salt and sugar dissolve in water but not in oil, because they dissolve only in certain solvents.

### Where students lose marks

**Trap 1 — Solute dissolves; solvent does the dissolving.** Salt (solute) dissolves in water (solvent).

**Trap 2 — For two liquids, amount decides the role.** Smaller amount = solute, larger = solvent — not which one "looks" like the liquid.

**Trap 3 — A solution is uniform.** Sand or chalk in water is non-uniform — not a true solution.

### Model answers — write them like this

**Example A. In salt water, name the solute and the solvent.**

**Step 1:** Salt dissolves → solute.

**Step 2:** Water does the dissolving → solvent.

**∴ Solute = salt; Solvent = water.**

**Example B. A little vinegar (acetic acid) is mixed with a lot of water. Which is the solute?**

**Step 1:** For two liquids, the smaller amount is the solute.

**$\therefore$  Acetic acid (smaller amount) is the solute; water is the solvent.**

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